Interactive Demo: Gelbrich / Dual

1. Compute **exact discrete** W_2 using POT (linear program).
2. Extract **dual potentials** and verify feasibility: $f_i + g_j \leq C_{ij}$
3. Implement the **Gelbrich bound** and check that Bound $\leq$ Exact

Hint for 1: check documentation for ot.emd

Hint for 2: broadcasting is your friend (but you can also use outer products)

For 3:

$$\|\mu_\alpha - \mu_\beta\|^2 + \mathscr{B}(\Sigma_\alpha, \Sigma_\beta)^2 \leq \mathrm{W}_2^2(\alpha, \beta)$$

$$\mathscr{B}(\Sigma_\alpha, \Sigma_\beta)^2 = \mathrm{tr}\Big(\Sigma_\alpha + \Sigma_\beta - 2\big(\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2}\big)^{1/2}\Big)$$

https://bit.ly/cs2840-lecture4

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